38

Diffraction Patterns and Polarization

CHAPTER OUTLINE

38.1 Introduction to Diffraction Patterns

38.2 Diffraction Patterns from Narrow Slits

38.3 Resolution of Single-Slit and Circular Apertures

38.4 The Diffraction Grating

38.5 Diffraction of X-Rays by Crystals

38.6 Polarization of Light Waves

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ38.1 Answer (a). Glare, as usually encountered when driving or boating, is horizontally polarized. Reflected light is polarized in the same plane as the reflecting surface. As unpolarized light hits a shiny horizontal surface, the atoms on the surface absorb and then reemit the light energy as a reflection. We can model the surface as containing conduction electrons free to vibrate easily along the surface, but not to move easily out of surface. The light emitted from a vibrating electron is partially or completely polarized along the plane of vibration, thus horizontally.

OQ38.2 Answer (c). The polarization state of a light beam that is reflected by a metallic surface is not changed; therefore, a beam of light that is not polarized before it is reflected is not polarized after it is reflected by a metallic surface.

OQ38.3 Answer (b). The wavelength will be much smaller than with visible light, so there will be no noticeable diffraction pattern.

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OQ38.4 Answer (b). In a single slit diffraction pattern, dark fringes occur where $\sin\theta_{\text{dark}} = m\lambda/a \approx \tan\theta_{\text{dark}} = y_{\text{dark}}/L$, and m is any non-zero integer.

Thus, the width of the slit, a, in the described situation, must be

$$a = \frac{m\lambda L}{(y_{\text{dark}})_1} = \frac{(1)\lambda L}{(y_{\text{dark}})_1} = \frac{(5.00\times10^{-7}\ \text{m})(1.00\ \text{m})}{5.00\times10^{-3}\ \text{m}}$$
$$= 1.00\times10^{-4}\ \text{m} = 0.100\ \text{mm}$$

OQ38.5 Answer (d). The central maximum lies between the first-order minima defined by the relation $\sin\theta_{\text{dark}} = m\lambda/a = \lambda/a$. Because the angle is small, $\sin\theta_{\text{dark}} \approx \tan\theta_{\text{dark}} = y_{\text{dark}}/L$, so the width of the central maximum is proportional to $L\lambda/a$. Thus, the central maximum becomes twice as wide if the slit width a becomes half as wide.

OQ38.6 The ranking is (e) > (c) >(a) > (b) > (d). The central maximum lies between the first-order minima defined by the relation $\sin\theta_{\text{dark}} = m\lambda/a = \lambda/a$. Because the angle is small, $\sin\theta_{\text{dark}} \approx \tan\theta_{\text{dark}} = y_{\text{dark}}/L$, so the width of the central maximum is proportional to $L\lambda/a$. We consider the value of $L\lambda/a$: (a) $L\lambda_0/a$, (b) $f = c/\lambda_0$, so for $f' = 3/2\,f$, $\lambda' = 2/3\,\lambda_0$, and the width is $L(2/3\,\lambda_0)/a = 2/3(L\lambda_0/a)$, (c) $L(1.5\lambda_0)/a = 32(L\lambda_0/a)$, (d) $L\lambda_0/(2a) = 1/2(L\lambda_0/a)$, (e) $(2L)\lambda_0/a = 2(L\lambda_0/a)$.

OQ38.7 Answer (b). From Malus' law, the intensity of the light transmitted through a polarizer (analyzer) having its transmission axis oriented at angle 45° to the plane of polarization of the incident polarized light is $I = I_{\max}\cos^2 45° = I_{\max}/2$. Therefore, the intensity passing through the second polarizer having its transmission axis oriented at angle $\theta = 90° - 45° = 45°$ is $I = (I_{\max}/2)\cos^2 45° = I_{\max}/4$.

OQ38.8 Answer (e). Diffraction of light as it passes through, or reflects from, the objective element of a telescope can cause the images of two sources having a small angular separation to overlap and fail to be seen as separate images. According to Equation 38.6, $\theta_{\min} = 1.22\,\lambda/D$, the minimum angular separation $\theta_{\min}$ two sources must have in order to be seen as separate sources is inversely proportional to the diameter D of the objective element. Thus, using a large-diameter objective element in a telescope increases its resolution.

OQ38.9 Answer (e). The bright colored patterns are the result of interference between light reflected from the upper surface of the oil and light reflected from the lower surface of the oil film.

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OQ38.10 Answer (b). No diffraction effects are observed because the separation distance between adjacent ribs is so much greater than the wavelength of x-rays. Diffraction does not limit the resolution of an x-ray image. Diffraction might sometimes limit the resolution of a sonogram.

OQ38.11 Answer (a). The grooves in a diffraction grating are not electrically conducting. Sending light through a diffraction grating is not like sending a vibration on a rope through a picket fence: there is no moving substance that could collide with the groove of the grating, so the grating could not prevent the wave from passing though it.

OQ38.12 Answer (c). The ability to resolve light sources depends on diffraction, not on intensity.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ38.1 The crystal cannot produce diffracted beams of visible light. The wavelengths of visible light are some hundreds of nanometers. There is no angle whose sine is greater than 1. Bragg's law, $2d \sin \theta = m\lambda$, cannot be satisfied for a wavelength much larger than the distance between atomic planes in the crystal.

CQ38.2 The wavelength of visible light is extremely small in comparison to the dimensions of your hand, so the diffraction of light around an obstacle the size of your hand is totally negligible. However, sound waves have wavelengths that are comparable to the dimensions of the hand or even larger. Therefore, significant diffraction of sound waves occurs around hand-sized obstacles.

CQ38.3 Since the obsidian is opaque, a standard method of measuring incidence and refraction angles and using Snell's Law is ineffective. Reflect unpolarized light from the horizontal surface of the obsidian through a vertically polarized filter. Change the angle of incidence until you observe that none of the reflected light is transmitted through the filter. This means that the reflected light is completely horizontally polarized, and that the incidence and reflection angles are the polarization angle. According to Equation 38.10, the tangent of the polarization angle is the index of refraction of the obsidian.

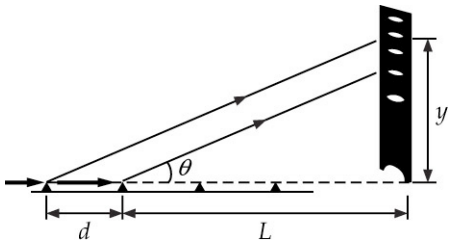
CQ38.4 (a) Light from the sky is partially polarized.

(b) Light from the blue sky that is polarized at 90° to the polarization axis of the glasses will be blocked, making the sky look darker as compared to the clouds.

CQ38.5 Consider incident light nearly parallel to the horizontal ruler. Suppose it scatters from bumps at distance d apart to produce a diffraction pattern on a vertical wall a distance L away. At a point of

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height y, where $\theta = \frac{y}{L}$ gives the scattering angle θ, the character of the interference is determined by the shift δ between beams scattered by adjacent bumps, where $\delta = d\cos\theta \approx d\left(1 + \frac{\theta^2}{2}\right)$. Bright spots appear for $\delta = m\lambda$, where $m = 1, 2, 3, \ldots$.



ANS. FIG. CQ38.5

For small θ, these equations combine and reduce to $m\lambda = d\left(1 + \frac{y_m^2}{2L^2}\right)$. Measurement of the heights y_m of bright spots allows calculation of the wavelength of the light. [Note that if a maximum occurs at $\theta = \frac{y}{L} \approx 0$, then scattered light from a bump constructively interferes with scattered light from the next bump in front, which constructively interferes with scattered light from the next bump...; thus $\lambda = d$.]

CQ38.6 First think about the glass without a coin and about one particular point P on the screen. We can divide up the area of the glass into ring-shaped zones centered on the line joining P and the light source, with successive zones contributing alternately in-phase and out-of-phase with the light that takes the straight-line path to P. These Fresnel zones have nearly equal areas. An outer zone contributes only slightly less to the total wave disturbance at P than does the central circular zone. Now insert the coin. If P is in line with its center, the coin will block off the light from some particular number of zones. The first unblocked zone around its circumference will send light to P with significant amplitude. Zones farther out will predominantly interfere destructively with each other, and the Arago spot is bright. Slightly off the axis there is nearly complete destructive interference, so most of the geometrical shadow is dark. A bug on the screen crawling out past the edge of the geometrical shadow would in effect see the central few zones coming out of eclipse. As the light from them interferes alternately constructively and destructively, the bug moves through bright and dark fringes on the screen. The diffraction pattern is shown in Figure 38.3 in the text.

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CQ38.7 The skin on the tip of a finger has a series of closely spaced ridges and swirls on it. When the finger touches a smooth surface, the oils from the skin will be deposited on the surface in the pattern of the closely spaced ridges. The clear spaces between the lines of deposited oil can serve as the slits in a crude diffraction grating and produce a colored spectrum of the light passing through or reflecting from the glass surface.

CQ38.8 (a) The diffraction pattern of a hair is the same as the diffraction pattern produced by a single slit of the same width.

(b) The central maximum is flanked by minima. Measure the width $2y$ of the central maximum between the minima bracketing it. Because the angle is small, you can use

$$\sin\theta_{\text{dark}} \approx \tan\theta_{\text{dark}}$$
$$m\lambda/a \approx y/L$$

to find the width a of the hair.

CQ38.9 The condition for constructive interference is that the three radio signals arrive at the city in phase. We know the speed of the waves (it is the speed of light c), the angular bearing θ of the city east of north from the broadcast site, and the distance d between adjacent towers. The wave from the westernmost tower must travel an extra distance $2d\sin\theta$ to reach the city, compared to the signal from the eastern tower. For each cycle of the carrier wave, the western antenna would transmit first, the center antenna after a time delay $\frac{d\sin\theta}{c}$, and the eastern antenna after an additional equal time delay.

CQ38.10 The correct orientation is vertical. If the horizontal width of the opening is equal to or less than the wavelength of the sound, then the equation $a\sin\theta = (1)\lambda$ has the solution $\theta = 90°$, or has no solution. The central diffraction maximum covers the whole seaward side. If the vertical height of the opening is large compared to the wavelength, then the angle in $a\sin\theta = (1)\lambda$ will be small, and the central diffraction maximum will form a thin horizontal sheet.

Featured in the motion picture *M*A*S*H* (20th Century Fox, Aspen Productions, 1970) is a loudspeaker mounted on an exterior wall of an Army barracks. It has an approximately rectangular aperture, and it is installed incorrectly. The longer side is horizontal, to maximize sound spreading in a vertical plane and to minimize sound radiated in different horizontal directions.

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CQ38.11 Audible sound has wavelengths on the order of meters or centimeters, while visible light has a wavelength on the order of half a micrometer. In this world of breadbox-sized objects, $\frac{\lambda}{a}$ is large for sound, and sound diffracts around walls with doorways. But $\frac{\lambda}{a}$ is a tiny fraction for visible light passing ordinary-size objects or apertures, so light changes its direction by only very small angles when it diffracts.

Another way of phrasing the answer: We can see by a small angle around a small obstacle or around the edge of a small opening. The side fringes in Figure 38.1 and the Arago spot in the center of Figure 38.3 show this diffraction. We cannot always hear around corners. Out-of-doors, away from reflecting surfaces, have someone a few meters distant face away from you and whisper. The high-frequency, short-wavelength, information-carrying components of the sound do not diffract around his head enough for you to understand his words.

Suppose an opera singer loses the tempo and cannot immediately get it from the orchestra conductor. Then the prompter may make rhythmic kissing noises with her lips and teeth. Try it—you will sound like a birdwatcher trying to lure out a curious bird. This sound is clear on the stage but does not diffract around the prompter's box enough for the audience to hear it.

CQ38.12 Consider vocal sound moving at 340 m/s and of frequency 3 000 Hz. Its wavelength is

$$\lambda = \frac{v}{f} = \frac{340 \text{ m/s}}{3000 \text{ Hz}} = 0.113 \text{ m}$$

If your mouth, for horizontal dispersion, behaves similarly to a slit 6.00 cm wide, then $a \sin\theta = m\lambda$ predicts no diffraction minima. You are a nearly isotropic source of this sound. It spreads out from you nearly equally in all directions. On the other hand, if you use a megaphone with width 60.0 cm at its wide end, then $a \sin\theta = m\lambda$ predicts the first diffraction minimum at

$$\theta = \sin^{-1}\left(\frac{m\lambda}{a}\right) = \sin^{-1}\left(\frac{0.113 \text{ m}}{0.600 \text{ m}}\right) = 10.9°$$

This suggests that the sound is radiated mostly toward the front into a diverging beam of angular diameter only about 20°. With less sound energy wasted in other directions, more is available for your intended auditors. We could check that a distant observer to the side or behind you receives less sound when a megaphone is used.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 38.2 Diffraction Patterns from Narrow Slits

***P38.1** (a) According to Equation 38.1, dark bands (minima) occur where

$$\sin\theta = m\frac{\lambda}{a}$$

For the first minimum, $m = 1$, and the distance from the center of the central maximum is

$$y_1 = L\tan\theta \approx L\sin\theta = L\left(\frac{\lambda}{a}\right)$$

Thus, the needed distance to the screen is

$$L = y_1\left(\frac{a}{\lambda}\right) = \left(0.85\times10^{-3}\text{ m}\right)\left(\frac{0.75\times10^{-3}\text{ m}}{587.5\times10^{-9}\text{ m}}\right) = \boxed{1.1\text{ m}}$$

(b) The width of the central maximum is

$$2y_1 = 2(0.85\text{ mm}) = \boxed{1.7\text{ mm}}$$

P38.2 From Equation 38.1, with $m = 1$,

$$\sin\theta = \frac{\lambda}{a} = \frac{6.328\times10^{-7}\text{ m}}{3.00\times10^{-4}\text{ m}} = 2.11\times10^{-3}$$

Then,

$$\frac{y}{1.00\text{ m}} = \tan\theta \approx \sin\theta \approx \theta \text{ (for small } \theta) \quad\rightarrow\quad y = 2.11\text{ mm}$$

and $2y = \boxed{4.22\text{ mm}}$

P38.3 If the speed of sound is 343 m/s,

$$\lambda = \frac{v}{f} = \frac{343\text{ m/s}}{650\text{ s}^{-1}} = 0.528\text{ m}$$

Diffraction minima occur at angles described by $a\sin\theta = m\lambda$.

$$(1.10\text{ m})\sin\theta_1 = 1(0.528\text{ m}) \quad \theta_1 = \pm28.7°$$

$$(1.10\text{ m})\sin\theta_2 = 2(0.528\text{ m}) \quad \theta_2 = \pm73.6°$$

$$(1.10\text{ m})\sin\theta_3 = 3(0.523\text{ m}) \quad \theta_3 \text{ nonexistent}$$

(a) There are $\boxed{\text{four}}$ minima.

(b) $\boxed{\theta = \pm28.7°, \pm73.6°}$

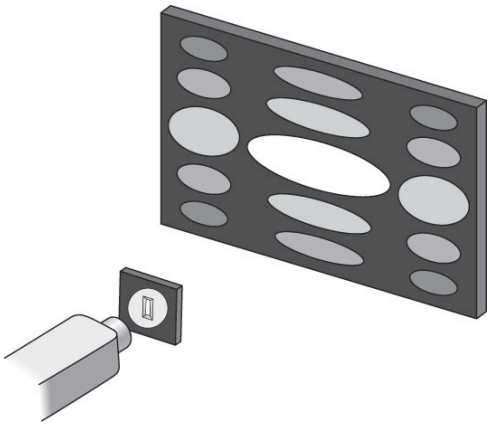
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P38.4 (a) Refer to ANS. FIG. P38.4. The rectangular patch on the wall is wider than it is tall. The aperture will be taller than it is wide. For horizontal spreading we have

$$\tan\theta_{\text{width}} = \frac{y_{\text{width}}}{L} = \frac{0.110 \text{ m}/2}{4.5 \text{ m}} = 0.012\,2$$

$$a_{\text{width}} \sin\theta_{\text{width}} = 1\lambda$$

$$a_{\text{width}} = \frac{632.8\times10^{-9} \text{ m}}{0.012\,2} = 5.18\times10^{-5} \text{ m} = \boxed{51.8\ \mu\text{m}}$$



ANS. FIG. P38.4

(b) For vertical spreading, similarly

$$\tan\theta_{\text{height}} = \frac{0.006 \text{ m}/2}{4.5 \text{ m}} = 0.000\,667$$

$$a_{\text{height}} = \frac{1\lambda}{\sin\theta_h} = \frac{632.8\times10^{-9} \text{ m}}{0.000\,667} = 9.49\times10^{-4} \text{ m} = \boxed{949\ \mu\text{m}}$$

(c) The longer dimension in the central bright patch is horizontal.

(d) The longer dimension of the aperture is vertical.

(e) A smaller distance between aperture edges causes a wider diffraction angle. The longer dimension of each rectangle is 18.3 times larger than the smaller dimension.

P38.5 For destructive interference, from Equation 38.1,

$$\sin\theta = m\frac{\lambda}{a} = \frac{\lambda}{a} = \frac{5.00 \text{ cm}}{36.0 \text{ cm}} = 0.139$$

and $\theta = 7.98°$. Then,

$$\frac{y}{L} = \tan\theta$$

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gives $y = L\tan\theta = (6.50\text{ m})\tan 7.98° = 0.912\text{ m}$

$= \boxed{91.2\text{ cm}}$

P38.6 In a single slit diffraction pattern, with the slit having width a, the dark fringe of order m occurs at angle θ_m, where $\sin\theta_m = m(\lambda/a)$ and $m = \pm 1,\ \pm 2,\ \pm 3, \ldots$. The location, on a screen located distance L from the slit, of the dark fringe of order m (measured from $y = 0$ at the center of the central maximum) is

$$(y_{\text{dark}})_m = L\tan\theta_m \approx L\sin\theta_m = m\lambda\left(\frac{L}{a}\right)$$

(a) The central maximum extends from the $m = +1$ dark fringe on one side to the $m = -1$ dark fringe on the other side, so the width of this central maximum is

$$\text{Central max. width} = (y_{\text{dark}})_{m=1} - (y_{\text{dark}})_{m=-1}$$
$$= (1)\left(\frac{\lambda L}{a}\right) - (-1)\left(\frac{\lambda L}{a}\right) = \frac{2\lambda L}{a}$$

Therefore,

$$L = \frac{a(\text{Central max. width})}{2\lambda}$$
$$= \frac{\left(0.200\times10^{-3}\text{ m}\right)\left(8.10\times10^{-3}\text{ m}\right)}{2\left(5.40\times10^{-7}\text{ m}\right)} = \boxed{1.50\text{ m}}$$

(b) The first order bright fringe extends from the $m = 1$ dark fringe to the $m = 2$ dark fringe, or

$$\left(\Delta y_{\text{bright}}\right)_1 = \left(y_{\text{dark}}\right)_{m=2} - \left(y_{\text{dark}}\right)_{m=1} = 2\left(\frac{\lambda L}{a}\right) - 1\left(\frac{\lambda L}{a}\right) = \frac{\lambda L}{a}$$
$$= \frac{\left(5.40\times10^{-7}\text{ m}\right)(1.50\text{ m})}{0.200\times10^{-3}\text{ m}}$$
$$= 4.05\times10^{-3}\text{ m} = \boxed{4.05\text{ mm}}$$

Note that the width of the first order bright fringe is exactly one half the width of the central maximum.

P38.7 In the equation for single-slit diffraction minima at small angles,

$$\frac{y}{L} \approx \sin\theta_{\text{dark}} = \frac{m\lambda}{a}$$

we take differences between the first and third dark fringes, to see that

$$\frac{\Delta y}{L} = \frac{\Delta m\lambda}{a} \quad \text{with} \quad \Delta y = 3.00\times10^{-3}\text{ m} \quad \text{and} \quad \Delta m = 3 - 1 = 2$$

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The width of the slit is then

$$a = \frac{\lambda L \Delta m}{\Delta y} = \frac{(690\times10^{-9}\ \text{m})(0.500\ \text{m})(2)}{(3.00\times10^{-3}\ \text{m})} = \boxed{2.30\times10^{-4}\ \text{m}}$$

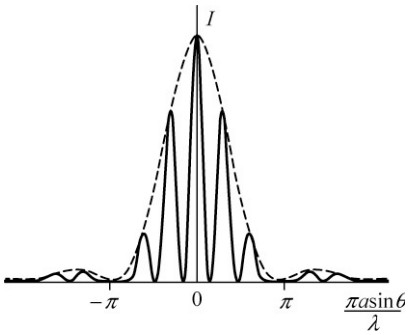
P38.8 Use the small-angle approximation: $\frac{y}{L} \approx \sin\theta = \frac{m\lambda}{a}$.

Then, $\frac{\Delta y}{L} = \left|\frac{m_2\lambda}{a} - \frac{m_1\lambda}{a}\right| = |m_2 - m_1|\frac{\lambda}{a} \rightarrow \boxed{a = \frac{\lambda L|m_2 - m_1|}{\Delta y}}$.

P38.9 The diffraction envelope shows a broad central maximum flanked by zeros at $a\sin\theta = 1\lambda$ and $a\sin\theta = 2\lambda$. That is, the zeros are at $(\pi a\sin\theta)/\lambda = \pi,\ -\pi,\ 2\pi,\ -2\pi, \ldots$ Noting that the distance between slits is $d = 9\ \mu\text{m} = 3a$, we say that within the diffraction envelope the interference pattern shows closely spaced maxima at $d\sin\theta = m\lambda$, giving $(\pi 3a\sin\theta)/\lambda = m\pi$ or

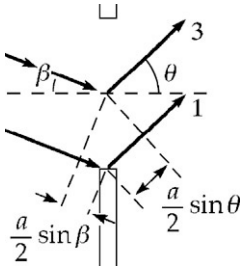
$$(\pi a\sin\theta)/\lambda = 0,\ \pi/3,\ -\pi/3,\ 2\pi/3,\ -2\pi/3$$

The third-order interference maxima are missing because they fall at the same directions as diffraction minima, but the fourth order can be visible at $(\pi a\sin\theta)/\lambda = 4\pi/3$ and $-4\pi/3$ as diagrammed.



ANS. FIG. P38.9

P38.10 Equation 38.1 states that $\sin\theta = \frac{m\lambda}{a}$, where $m = \pm1,\ \pm2,\ \pm3, \ldots$. The requirement for $m = 1$ is from an analysis of the extra path distance traveled by ray 1 compared to ray 3 in the textbook Figure 38.5. This extra distance must be equal to $\frac{\lambda}{2}$ for destructive interference. When the source rays approach the slit at an angle β, there is a distance added to the path difference (of ray 1 compared to



ANS. FIG. P38.10

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ray 3) of $\frac{a}{2}\sin\beta$. Then, for destructive interference,

$$\frac{a}{2}\sin\beta + \frac{a}{2}\sin\theta = \frac{\lambda}{2} \quad \text{so} \quad \sin\theta = \frac{\lambda}{a} - \sin\beta$$

Dividing the slit into 4 parts leads to the second order minimum:

$$\frac{a}{4}\sin\beta + \frac{a}{4}\sin\theta = \frac{\lambda}{2} \quad \text{so} \quad \sin\theta = \frac{2\lambda}{a} - \sin\beta$$

Dividing the slit into 6 parts gives the third order minimum:

$$\sin\theta = \frac{3\lambda}{a} - \sin\beta$$

Generalizing, we obtain the condition for the mth order minimum:

$$\sin\theta = \frac{m\lambda}{a} - \sin\beta \qquad m = \pm 1,\ \pm 2,\ \pm 3,\ \ldots$$

P38.11 First we find where we are. The angle to the side is small so

$$\sin\theta \approx \tan\theta = \frac{y}{L} = \frac{4.10 \times 10^{-3}\ \text{m}}{1.20\ \text{m}} = 3.417 \times 10^{-3}$$

The parameter controlling the intensity is

$$\frac{\pi a \sin\theta}{\lambda} = \frac{\pi\left(4.00 \times 10^{-4}\ \text{m}\right)\left(3.417 \times 10^{-3}\right)}{546.1 \times 10^{-9}\ \text{m}} = 7.862\ \text{rad}$$

This is between 2π and 3π, so the point analyzed is off in the second side fringe. The fractional intensity is

$$\frac{I}{I_{\max}} = \left[\frac{\sin\left(\pi a \sin\theta/\lambda\right)}{\pi a \sin\theta/\lambda}\right]^2 = \left[\frac{\sin(7.862\ \text{rad})}{7.862\ \text{rad}}\right]^2 = \boxed{1.62 \times 10^{-2}}$$

P38.12 (a) Double-slit interference maxima are at angles given by $d\sin\theta = m\lambda$.

For $m = 0$, $\theta_0 = \boxed{0°}$

For $m = 1$, $(2.80\ \mu\text{m})\sin\theta = 1(0.501\,5\ \mu\text{m})$:

$\theta_1 = \sin^{-1}(0.179) = \boxed{\pm 10.3°}$

Similarly, for $m = 2$, 3, 4, and 5,

$\theta_2 = \boxed{\pm 21.0°}$, $\theta_3 = \boxed{\pm 32.5°}$, $\theta_4 = \boxed{\pm 45.8°}$, and $\theta_5 = \boxed{\pm 63.6°}$

For $m > 5$, there are no maxima.

(b) Thus, there are $5 + 5 + 1 = \boxed{11}$ directions for interference maxima.

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(c) We check for missing orders by looking for single-slit diffraction minima, at $a\sin\theta = m\lambda$.

For $m = 1$, $(0.700\ \mu\text{m})\sin\theta = 1(0.501\,5\ \mu\text{m})$ and $\theta_1 = \boxed{\pm 45.8^\circ}$

Thus, there is no bright fringe at this angle.

(d) From our answer to (c), $\boxed{\text{two}}$.

(e) $\boxed{\text{two}}$

(f) Two are missing because slit-slit minimum occur where a double-slit maximum would be: $\boxed{\text{nine}}$

(g) $$I = I_{\max}\left[\frac{\sin(\pi a\sin\theta/\lambda)}{\pi\sin\theta/\lambda}\right]^2$$

At $\theta = 63.6^\circ$,

$$\frac{\pi a\sin\theta}{\lambda} = \frac{\pi(0.700\ \mu\text{m})\sin 63.6^\circ}{(0.501\,5\ \mu\text{m})} = 3.93\ \text{rad} = 225^\circ$$

and $I = \boxed{0.032\,4 I_{\max}}$

P38.13 With the screen locations of the dark fringe of order m at

$$(y_{\text{dark}})_m = L\tan\theta_m \approx L\sin\theta_m = m(\lambda L/a) \quad \text{for } m = \pm 1, \pm 2, \pm 3, \ldots$$

the width of the central maximum is

$$\Delta y_{\substack{\text{central}\\ \text{maximum}}} = (y_{\text{dark}})_{m=+1} - (y_{\text{dark}})_{m=-1} = 2(\lambda L/a)$$

so

$$\lambda = \frac{a\left(\Delta y_{\substack{\text{central}\\ \text{maximum}}}\right)}{2L} = \frac{(0.600\times 10^{-3}\ \text{m})(2.00\times 10^{-3}\ \text{m})}{2(1.30\ \text{m})}$$

$$= 4.62\times 10^{-7}\ \text{m} = \boxed{462\ \text{nm}}$$

Section 38.3 Resolution of Single-Slit and Circular Apertures

P38.14 We assume Rayleigh's criterion applies to the cat's eye with pupil narrowed. For a single slit (not a round aperture), for small angles

$$\theta \approx \sin\theta = \frac{\lambda}{a} = \frac{500\times 10^{-9}\ \text{m}}{0.500\times 10^{-3}\ \text{m}} = \boxed{1.00\times 10^{-3}\ \text{rad}}$$

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P38.15 Using Rayleigh's criterion,

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 0.100°\left(\frac{\pi \text{ rad}}{180°}\right) = 1.75\times10^{-3} \text{ rad}$$

and $$D = 1.22\left(\frac{\lambda}{\theta_{\min}}\right) = 1.22\left(\frac{3.00\times10^{-3} \text{ m}}{\theta_{\min}}\right) = \boxed{2.10 \text{ m}}$$

P38.16 Using Rayleigh's criterion, $\theta_{\min} = 1.22\frac{\lambda}{D} = \frac{y}{L}$. Therefore,

$$L = \frac{yD}{1.22\lambda} = \frac{\left(2.80\times10^{-2} \text{ m}\right)\left(0.600\times10^{-3} \text{ m}\right)}{1.22\left(550\times10^{-9} \text{ m}\right)} = \boxed{25.0 \text{ m}}$$

P38.17 Using Rayleigh's criterion, $\theta_{\min} = 1.22\frac{\lambda}{D} = \frac{y}{L}$. Therefore,

$$y = 1.22\left(\frac{\lambda}{D}\right)L = 1.22\left(\frac{500\times10^{-9} \text{ m}}{58.0\times10^{-2} \text{ m}}\right)\left(270\times10^{3} \text{ m}\right)$$

$$= \boxed{0.284 \text{ m}}$$

P38.18 (a) The limiting angle for the resolution of the microscope is

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{589\times10^{-9} \text{ m}}{9.00\times10^{-3} \text{ m}}\right) = 7.98\times10^{-5} \text{ rad}$$

$$= \boxed{79.8\ \mu\text{rad}}$$

(b) For a smaller angle of diffraction we choose the smallest visible wavelength, violet at 400 nm, to obtain

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{400\times10^{-9} \text{ m}}{9.00\times10^{-3} \text{ m}}\right) = 5.42\times10^{-5} \text{ rad}$$

$$= \boxed{54.2\ \mu\text{rad}}$$

(c) The wavelength in water is shortened to its vacuum value divided by the index of refraction. The resolving power is improved, with the minimum resolvable angle becoming

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{589\times10^{-9} \text{ m}/1.33}{9.00\times10^{-3} \text{ m}}\right) = 6.00\times10^{-5} \text{ rad}$$

$$= \boxed{60.0\ \mu\text{rad}}$$

Better than water for many purposes is oil immersion.

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P38.19 When the pupil is open wide, it appears that the resolving power of human vision is limited by the coarseness of light sensors on the retina. But we use Rayleigh's criterion as a handy indicator of how good our vision might be. We are given

$$L = 250 \times 10^3 \text{ m},\ \lambda = 5.00 \times 10^{-7} \text{ m, and } d = 5.00 \times 10^{-3} \text{ m}$$

The smallest object the astronauts can resolve is given by Rayleigh's criterion, $\theta_{\min} = 1.22\dfrac{\lambda}{D} = \dfrac{y}{L}$. Therefore,

$$y = 1.22\frac{\lambda}{D}L = 1.22\left(\frac{5.00 \times 10^{-7} \text{ m}}{5.00 \times 10^{-3} \text{ m}}\right)\left(250 \times 10^3 \text{ m}\right) = \boxed{30.5 \text{ m}}$$

P38.20 Undergoing diffraction from a circular opening, the beam spreads into a cone of half-angle

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{632.8 \times 10^{-9} \text{ m}}{0.005\,00 \text{ m}}\right) = 1.54 \times 10^{-4} \text{ rad}$$

The radius of the beam ten kilometers away is, from the definition of radian measure,

$$r_{\text{beam}} = \theta_{\min}\left(1.00 \times 10^4 \text{ m}\right) = 1.544 \text{ m}$$

and its diameter is $d_{\text{beam}} = 2r_{\text{beam}} = \boxed{3.09 \text{ m}}$.

***P38.21** The limit of resolution in air is

$$\theta_{\min}\big|_{\text{air}} = 1.22\frac{\lambda}{D} = 0.60\ \mu\text{rad}$$

In oil, the limiting angle of resolution will be

$$\theta_{\min}\big|_{\text{oil}} = 1.22\frac{\lambda_{\text{oil}}}{D} = 1.22\frac{\left(\lambda/n_{\text{oil}}\right)}{D} = \frac{1}{n_{\text{oil}}}\left(1.22\frac{\lambda}{D}\right)$$

or $$\theta_{\min}\big|_{\text{oil}} = \frac{\theta_{\min}\big|_{\text{air}}}{n_{\text{oil}}} = \frac{0.60\ \mu\text{rad}}{1.5} = \boxed{0.40\ \mu\text{rad}}$$

P38.22 When the pupil is open wide, it appears that the resolving power of human vision is limited by the coarseness of light sensors on the retina. But we use Rayleigh's criterion as a handy indicator of how good our vision might be. We take $\theta_{\min} = \dfrac{d}{L} = 1.22\dfrac{\lambda}{D}$, where $\theta_{\min}$ is the smallest angular separation of two objects for which they are resolved by an aperture of diameter D, d is the separation of the two objects, and L is the maximum distance of the aperture from the two objects at which they can be resolved.

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(a) Two objects can be resolved if their angular separation is greater than $\theta_{\min}$. Thus, $\theta_{\min}$ should be as small as possible. Therefore, light with the smaller of the two given wavelengths is easier to resolve, i.e., blue.

(b) $$L = \frac{Dd}{1.22\lambda} = \frac{\left(5.20\times10^{-3}\text{ m}\right)\left(2.80\times10^{-2}\text{ m}\right)}{1.22\lambda} = \frac{1.193\times10^{-4}\text{ m}^2}{\lambda}$$

Thus for $\lambda = 640$ nm, $L = 186$ m, and for $\lambda = 440$ nm, $L = 271$ m. The viewer with the assumed diffraction-limited vision could resolve adjacent tubes of blue in the range $\boxed{186\text{ m to }271\text{ m}}$, but cannot resolve adjacent tubes of red in this range.

P38.23 When the pupil is open wide, it appears that the resolving power of human vision is limited by the coarseness of light sensors on the retina. But we use Rayleigh's criterion as a handy indicator of how good our vision might be. According to this criterion, two dots separated center-to-center by 2.00 mm would overlap when

$$\theta_{\min} = \frac{d}{L} = 1.22\frac{\lambda}{D}$$

Thus, $$L = \frac{dD}{1.22\lambda} = \frac{\left(2.00\times10^{-3}\text{ m}\right)\left(5.00\times10^{-3}\text{ m}\right)}{1.22\left(500\times10^{-9}\text{ m}\right)} = \boxed{16.4\text{ m}}.$$

P38.24 We are given $D = 2.10$ m and $L = 9\,000$ m. The wavelength of the Coast Guard radar is

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\text{ m/s}}{15.0\times10^{9}\text{ Hz}} = 0.020\,0\text{ m}$$

From Rayleigh's criterion, $\theta_{\min} = 1.22\frac{\lambda}{D} = \frac{d}{L}$. Therefore,

$$d = 1.22\left[\frac{(0.020\,0\text{ m})(9\,000\text{ m})}{2.10\text{ m}}\right] = \boxed{105\text{ m}}$$

Section 38.4 The Diffraction Grating

P38.25 The first order maximum occurs at 20.5°, so $\sin\theta = \sin 20.5° = 0.350$, and, from Equation 38.7,

$$d = \frac{\lambda}{\sin\theta} = \frac{632.8\text{ nm}}{0.350} = 1.81\times10^{3}\text{ nm}$$

Therefore, the line spacing = $\boxed{1.81\ \mu\text{m}}$

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P38.26 The ruling engine that cut the diffraction grating (or the aluminum plate from which the gelatin or plastic was cast) sliced each centimeter into two thousand divisions. So the grating spacing is

$$d = \frac{1.00 \times 10^{-2}\ \text{m}}{2\ 000} = 5.00 \times 10^{-6}\ \text{m}$$

The light is deflected according to $d \sin\theta = m\lambda$:

$$\theta = \sin^{-1}\left(\frac{m\lambda}{d}\right) = \sin^{-1}\left[\frac{1\left(640 \times 10^{-9}\ \text{m}\right)}{5.00 \times 10^{-6}\ \text{m}}\right] = \boxed{7.35°}$$

P38.27 The sound has wavelength $\lambda = \frac{v}{f} = \frac{343\ \text{m/s}}{37.2 \times 10^{3}/\text{s}} = 9.22 \times 10^{-3}\ \text{m}$. Each diffracted beam is described by $d \sin\theta = m\lambda, \quad m = 0, 1, 2, \ldots$

The zero-order beam is at $m = 0$, $\theta = 0$. The beams in the first order of interference are to the left and right at

$$\theta = \sin^{-1}\left(\frac{1\lambda}{d}\right) = \sin^{-1}\left(\frac{9.22 \times 10^{-3}\ \text{m}}{1.30 \times 10^{-2}\ \text{m}}\right) = \sin^{-1} 0.709 = 45.2°$$

For a second-order beam we would need

$$\theta = \sin^{-1}\left(\frac{2\lambda}{d}\right) = \sin^{-1}\left[2(0.709)\right] = \sin^{-1}(1.42)$$

No angle, smaller or larger than 90°, has a sine greater than 1. Then a diffracted beam does not exist for the second order or any higher order. The whole answer is then:

(a) There are $\boxed{\text{three}}$ beams.

(b) The beams are at $\boxed{0°, +45.2°, -45.2°}$.

P38.28 (a) $d = \frac{10^{-2}\ \text{m}}{3\ 660} = 2.732 \times 10^{-6}\ \text{m} = 2\ 732\ \text{nm}$

$\lambda = \frac{d \sin\theta}{m}$, and $m = 1$: At $\theta = 10.1°$, $\lambda = \boxed{479\ \text{nm}}$

At $\theta = 13.7°$, $\lambda = \boxed{647\ \text{nm}}$.

At $\theta = 14.8°$, $\lambda = \boxed{698\ \text{nm}}$.

(b) $d = \frac{\lambda}{\sin\theta_1}$ and $2\lambda = d \sin\theta_2$ so $\sin\theta_2 = \frac{2\lambda}{d} = \frac{2\lambda}{\lambda/\sin\theta_1} = 2\sin\theta_1$.

Therefore, if $\theta_1 = 10.1°$ then $\sin\theta_2 = 2\sin(10.1°)$ gives $\theta_2 = \boxed{20.5°}$.

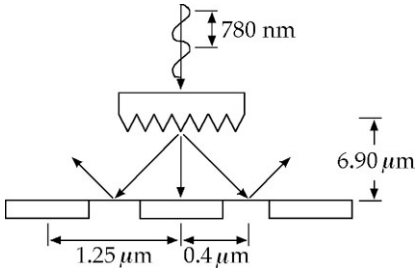
Similarly, for $\theta_1 = 13.7°$, $\theta_2 = \boxed{28.3°}$ and for $\theta_1 = 14.8°$, $\theta_2 = \boxed{30.7°}$.

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P38.29 For a side maximum, $\tan\theta = \frac{y}{L} = \frac{0.400\ \mu\text{m}}{6.90\ \mu\text{m}}$, which gives $\theta = 3.32°$.

Then, from $d\sin\theta = m\lambda$,

$$d = \frac{(1)\left(780\times10^{-9}\ \text{m}\right)}{\sin 3.32°} = 13.5\ \mu\text{m}$$



ANS. FIG. P38.29

The number of grooves per millimeter

$$= \frac{1\times10^{-3}\ \text{m}}{13.5\times10^{-6}\ \text{m}} = \boxed{74.2}$$

P38.30 The grating spacing is

$$d = \frac{1.00\times10^{-3}\ \text{m}}{250} = 4.00\times10^{-6}\ \text{m} = 4\ 000\ \text{nm}$$

Solving for m in Equation 38.7 gives

$$d\sin\theta = m\lambda \quad \rightarrow \quad m = \frac{d\sin\theta}{\lambda}$$

(a) The number of times a complete order is seen is the same as the number of orders in which the long wavelength limit is visible.

$$m_{\max} = \frac{d\sin\theta_{\max}}{\lambda} = \frac{(4\ 000\ \text{nm})\sin 90.0°}{700\ \text{nm}} = 5.71$$

or $\boxed{\text{5 orders is the maximum}}$.

(b) The highest order in which the violet end of the spectrum can be seen is:

$$m_{\max} = \frac{d\sin\theta_{\max}}{\lambda} = \frac{(4\ 000\ \text{nm})\sin 90.0°}{400\ \text{nm}} = 10.0$$

or $\boxed{\text{10 orders in the short-wavelength region}}$.

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P38.31 The grating spacing is

$$d = \frac{1.00 \times 10^{-2}\ \text{m}}{4200} = 2.38 \times 10^{-6}\ \text{m} = 2380\ \text{nm}$$

Solving for the angle θ from Equation 38.7, $d \sin\theta = m\lambda$, gives

$$\theta = \sin^{-1}\left(\frac{m\lambda}{d}\right)$$

Then, $$y = L\tan\theta = L\tan\left[\sin^{-1}\left(\frac{m\lambda}{d}\right)\right]$$

Thus, $$\Delta y = L\left\{\tan\left[\sin^{-1}\left(\frac{m\lambda_2}{d}\right)\right] - \tan\left[\sin^{-1}\left(\frac{m\lambda_1}{d}\right)\right]\right\}$$

For $m = 1$,

$$\Delta y = (2.00\ \text{m})\left\{\tan\left[\sin^{-1}\left(\frac{589.6}{2380}\right)\right] - \tan\left[\sin^{-1}\left(\frac{589}{2380}\right)\right]\right\}$$

$$= 0.554\ \text{mm}$$

For $m = 2$,

$$\Delta y = (2.00\ \text{m})\left\{\tan\left[\sin^{-1}\left(\frac{2(589.6)}{2380}\right)\right] - \tan\left[\sin^{-1}\left(\frac{2(589)}{2380}\right)\right]\right\}$$

$$= 1.54\ \text{mm}$$

For $m = 3$,

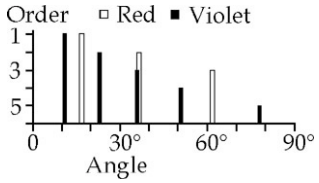
$$\Delta y = (2.00\ \text{m})\left\{\tan\left[\sin^{-1}\left(\frac{3(589.6)}{2380}\right)\right] - \tan\left[\sin^{-1}\left(\frac{3(589)}{2380}\right)\right]\right\}$$

$$= 5.04\ \text{mm}$$

Thus, the observed order must be $\boxed{m = 2}$.

P38.32 The grating spacing is

$$d = \frac{1.00 \times 10^{-2}\ \text{m}}{4\,500} = 2.22 \times 10^{-6}\ \text{m}$$



ANS. FIG. P38.32

In the 1st-order spectrum, diffraction angles are given by

$$\sin\theta = \frac{\lambda}{d}: \qquad \sin\theta_1 = \frac{656 \times 10^{-9}\ \text{m}}{2.22 \times 10^{-6}\ \text{m}} = 0.295$$

so that for red $\theta_1 = 17.17°$

and for blue $$\sin\theta_2 = \frac{434 \times 10^{-9}\ \text{m}}{2.22 \times 10^{-6}\ \text{m}} = 0.195,$$

so that $\theta_2 = 11.26°$.

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The angular separation is in first-order,

$$\Delta\theta = 17.17° - 11.26° = \boxed{5.91°}$$

In the second-order spectrum,

$$\Delta\theta = \sin^{-1}\left(\frac{2\lambda_1}{d}\right) - \sin^{-1}\left(\frac{2\lambda_2}{d}\right) = \boxed{13.2°}$$

Again, in the third order,

$$\Delta\theta = \sin^{-1}\left(\frac{3\lambda_1}{d}\right) - \sin^{-1}\left(\frac{3\lambda_2}{d}\right) = \boxed{26.5°}$$

Since the red does not appear in the fourth-order spectrum, the answer is complete.

P38.33 The principal maxima are defined by $d\sin\theta = m\lambda$, where $m = 0, 1, 2, \ldots$

For $m = 1$, $\lambda = d\sin\theta$.

Here, θ is the angle between the central ($m = 0$) and the first order ($m = 1$) maxima. The value of θ can be determined from the information given about the distance between maxima and the grating-to-screen distance:

$$\tan\theta = \frac{0.488 \text{ m}}{1.72 \text{ m}} = 0.284$$

so $\theta = 15.8°$ and $\sin\theta = 0.273$.

The distance between grating "slits" equals the reciprocal of the number of grating lines per centimeter

$$d = \frac{1}{5\,310 \text{ cm}^{-1}} = 1.88 \times 10^{-4} \text{ cm} = 1.88 \times 10^{3} \text{ nm}$$

The wavelength is

$$\lambda = d\sin\theta = \left(1.88 \times 10^{3} \text{ nm}\right)(0.273) = \boxed{514 \text{ nm}}$$

P38.34 From Equation 38.7, $\sin\theta = \dfrac{m\lambda}{d}$

Therefore, taking the ends of the visible spectrum to be $\lambda_v = 400$ nm and $\lambda_r = 750$ nm, the ends of the different order spectra are defined by:

End of second order: $\sin\theta_{2r} = \dfrac{2\lambda_r}{d} = \dfrac{1\,500 \text{ nm}}{d}$

Start of third order: $\sin\theta_{3v} = \dfrac{3\lambda_v}{d} = \dfrac{1\,200 \text{ nm}}{d}$

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Thus, it is seen that $\boxed{\theta_{2r} > \theta_{3v} \text{ and these orders must overlap}}$ regardless of the value of the grating spacing d.

P38.35 (a) We use the grating equation $d\sin\theta = m\lambda$:

$$d = \frac{m\lambda}{\sin\theta} = \frac{3\left(5.00\times10^{-7}\ \text{m}\right)}{\sin 32.0^\circ} = 2.83\times10^{-6}\ \text{m}$$

Thus the grating gauge is

$$\frac{1}{d} = 3.53\times10^{5}\ \text{grooves/m} = \boxed{3.53\times10^{3}\ \text{grooves/cm}}$$

(b) For any interference maximum for this light going through this grating,

$$\sin\theta = m\left(\frac{\lambda}{d}\right) = \frac{m\left(5.00\times10^{-7}\ \text{m}\right)}{2.83\times10^{-6}\ \text{m}} = m(0.177)$$

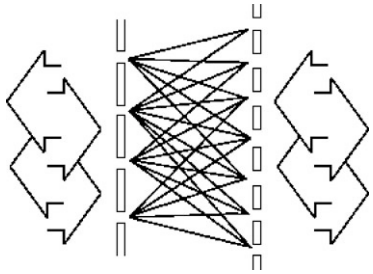
For $\sin\theta \le 1$, we require that $m(0.177) \le 1$ or $m \le 5.66$. Because m must be an integer, its maximum value is really 5. Therefore, the total number of maxima is $2m+1 = \boxed{11}$.

P38.36 (a) The several narrow parallel slits make a diffraction grating. The zeroth- and first-order maxima are separated according to

$$d\sin\theta = (1)\lambda \qquad \sin\theta = \frac{\lambda}{d} = \frac{632.8\times10^{-9}\ \text{m}}{1.2\times10^{-3}\ \text{m}}$$

$$\theta = \sin^{-1}\left(0.000\,527\right) = 0.000\,527\ \text{rad}$$

$$y = L\tan\theta = \left(1.40\ \text{m}\right)\left(0.000\,527\right) = \boxed{0.738\ \text{mm}}$$



ANS. FIG. P38.36

(b) Many equally spaced transparent lines appear on the film. It is itself a diffraction grating. When the same light is sent through the film, it produces interference maxima separated according to

$$d\sin\theta = (1)\lambda \quad \rightarrow \quad \sin\theta = \frac{\lambda}{d} = \frac{632.8\times10^{-9}\ \text{m}}{0.738\times10^{-3}\ \text{m}} = 0.000\,857$$

$$y = L\tan\theta = \left(1.40\ \text{m}\right)\left(0.000\,857\right) = 1.20\ \text{mm}$$

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An image of the original set of slits appears on the screen. If the screen is removed, light diverges from the real images with the same wave fronts reconstructed as the original slits produced. Reasoning from the mathematics of Fourier transforms, Gabor showed that light diverging from any object, not just a set of slits, could be used. In the picture, the slits or maxima on the left are separated by 1.20 mm. The slits or maxima on the right are separated by 0.738 mm. The length difference between any pair of lines is an integer number of wavelengths. Light can be sent through equally well toward the right or toward the left.

P38.37 Fifteen bright spots means that the central maximum and seven orders of side maxima appear.

(a) If the seventh order is at less than 90°, the eighth order might be nearly ready to appear according to

$$d\sin\theta = m\lambda$$

$$d(1) = 8\left(654\times10^{-9}\text{ m}\right) \rightarrow d = 5.23\ \mu\text{m}$$

$$d = 5.23\times10^{-6}\text{ m} = \boxed{5.23\ \mu\text{m}}$$

(b) If the seventh order is just at 90°,

$$d\sin\theta = m\lambda$$

$$d(1) = 7\left(654\times10^{-9}\text{ m}\right)$$

$$d = 4.58\times10^{-6}\text{ m} = \boxed{4.58\ \mu\text{m}}$$

Section 38.5 Diffraction of X-Rays by Crystals

P38.38 The atomic planes in this crystal are shown in Figure 38.22 of the text. The diffraction they produce is described by Bragg's law,

$$2d\sin\theta = m\lambda: \qquad \sin\theta = \frac{m\lambda}{2d} = \frac{1\left(0.140\times10^{-9}\text{ m}\right)}{2\left(0.281\times10^{-9}\text{ m}\right)} = 0.249$$

and $\boxed{\theta = 14.4^\circ}$.

P38.39 The grazing angle is measured from the surface, as shown in Figure 38.23. Then, from $2d\sin\theta = m\lambda$,

$$\lambda = \frac{2d\sin\theta}{m}$$

$$= \frac{2\left(0.353\times10^{-9}\text{ m}\right)\sin 7.60^\circ}{1} = 9.34\times10^{-11}\text{ m} = \boxed{0.093\ 4\text{ nm}}$$

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P38.40 From $2d\sin\theta = m\lambda$,

$$\sin\theta = \frac{m\lambda}{2d} = \frac{2(0.166\text{ nm})}{2(0.314\text{ nm})} = 0.529$$

and $\boxed{\theta = 31.9°}$

P38.41 (a) By Bragg's law, $2d\sin\theta = m\lambda$, and $m = 2$:

$$\lambda = 2d\sin\theta = 2(0.250\text{ nm})\sin 12.6° = \boxed{0.109\text{ nm}}$$

(b) We obtain the number of orders from

$$\frac{m\lambda}{2d} = \sin\theta \le 1 \;\rightarrow\; m \le \frac{2d}{\lambda} = \frac{2(0.250\text{ nm})}{0.109\text{ nm}} = 4.59$$

The order-number must be an integer, so the largest value m can have is 4: $\boxed{\text{four}}$ orders can be observed.

Section 38.6 Polarization of Light Waves

P38.42 In Equation 38.10, $\tan\theta_p = n_2/n_1$, the index of refraction n_2 of the solid material must be larger than that of air ($n_1 = 1.00$). Therefore, we must have $\tan\theta_p > 1$. For this to be true, we must have $\theta_p > 45°$, so $\theta_p = 41.0°$ is not possible.

P38.43 We define the initial angle, at which all the light is transmitted, to be $\theta = 0$. Turning the disk to another angle will then reduce the transmitted light by an intensity factor as described by $I = I_{\max}\cos^2\theta$.

Then, $\theta = \cos^{-1}\sqrt{\dfrac{I}{I_{\max}}}$

(a) For $I = I_{\max}/3.00$,

$$\theta = \cos^{-1}\left(\frac{I}{I_{\max}}\right)^{1/2} = \cos^{-1}\frac{1}{\sqrt{3.00}} = \cos^{-1}0.577 = \boxed{54.7°}$$

(b) Now $\theta = \cos^{-1}\left(\dfrac{I}{I_{\max}}\right)^{1/2} = \cos^{-1}\dfrac{1}{\sqrt{5.00}} = \cos^{-1}0.447 = \boxed{63.4°}$

(c) The largest factor of intensity reduction requires the largest crossing angle,

$$\theta = \cos^{-1}\left(\frac{I}{I_{\max}}\right)^{1/2} = \cos^{-1}\frac{1}{\sqrt{10.0}} = \cos^{-1}0.316 = \boxed{71.6°}$$

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P38.44 By Brewster's law, for light in air ($n = 1.00$) reflecting off a surface of index n,

$$\tan\theta_p = \frac{n_2}{n_1} = \frac{n}{1.00} = n$$

$$n = \tan\theta_p = \tan(48.0°) = \boxed{1.11}$$

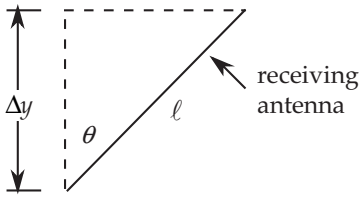
P38.45 The intensity of unpolarized light passing through the first polarizing filter is reduced by 1/2. The second transmits $\cos^2 30.0° = \frac{3}{4}$.

$$\frac{I}{I_{\max}} = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8} = \boxed{0.375}$$

***P38.46** $P = \dfrac{(\Delta V)^2}{R}$ or $P \propto (\Delta V)^2$

$\Delta V = (-)E_y \cdot \Delta y = E_y \cdot \ell \cos\theta$

$\Delta V \propto \cos\theta$ so $P \propto \cos^2\theta$



ANS. FIG. P38.46

(a) $\theta = 15.0°$:

$$P = P_{\max}\cos^2(15.0°) = 0.933P_{\max} = \boxed{93.3\%}$$

(b) $\theta = 45.0°$: $P = P_{\max}\cos^2(45.0°) = 0.500P_{\max} = \boxed{50.0\%}$

(c) $\theta = 90.0°$: $P = P_{\max}\cos^2(90.0°) = \boxed{0.00\%}$

P38.47 Let the first sheet have its axis at angle θ to the original plane of polarization, and let each further sheet have its axis turned by the same angle.

The first sheet passes intensity $I_{\max}\cos^2\theta$,

The second sheet passes $\left(I_{\max}\cos^2\theta\right)\cos^2\theta = I_{\max}\cos^4\theta$,

and the nth sheet lets through $I_{\max}\cos^{2n}\theta \geq 0.90I_{\max}$, where $\theta = \dfrac{45°}{n}$.

Try different integers to find n; for example,

$$\cos^{2\times5}\left(\frac{45°}{5}\right) = 0.885, \quad \cos^{2\times6}\left(\frac{45°}{6}\right) = 0.902, \quad \cos^{2\times7}\left(\frac{45°}{7}\right) = 0.915$$

(a) So $n = \boxed{6}$.

(b) $\theta = \dfrac{45°}{6} = \boxed{7.50°}$

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P38.48 (a) Let I_0 represent the intensity of unpolarized light incident on the first polarizer. The intensity of unpolarized light passing through a polarizing filter is reduced by 1/2, so the first filter lets through 1/2 of the incident intensity. Of the light reaching them, the second filter passes $\cos^2 45° = 1/2$ and the third filter also $\cos^2 45° = 1/2$. The transmitted intensity is then

$$I_0\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 0.125I_0$$

The reduction in intensity is by a factor of 1.00 – 0.125 = $\boxed{0.875}$ of the incident intensity.

(b) By the same logic as in part (a) we have transmitted

$$I_0\left(\frac{1}{2}\right)\left(\cos^2 30.0°\right)\left(\cos^2 30.0°\right)\left(\cos^2 30.0°\right) = \left(\frac{I_0}{2}\right)\left(\cos^2 30.0°\right)^3$$
$$= 0.211I_0$$

Then the fraction absorbed is 1.00 – 0.211 = $\boxed{0.789}$.

(c) Yet again we compute transmission

$$I_0\left(\frac{1}{2}\right)\left(\cos^2 15.0°\right)^6 = 0.330I_0$$

And the fraction absorbed is 1.00 – 0.330 = $\boxed{0.670}$.

(d) $\boxed{\text{We can get more and more of the incident light through the stack of ideal filters, approaching 50\%, by reducing the angle between the transmission axes of each one and the next.}}$

P38.49 For the polarizing angle,

$$\frac{n_{\text{sapphire}}}{n_{\text{air}}} = \tan\theta_p \quad \text{and} \quad \theta_p = \tan^{-1}\left(\frac{n_{\text{sapphire}}}{1.00}\right)$$

For the critical angle for total internal reflection,

$$n_{\text{sapphire}} \sin\theta_c = n_{\text{air}} \sin 90° = 1.00 \quad \text{so} \quad n_{\text{sapphire}} = \frac{1}{\sin\theta_c}$$

Therefore,

$$\theta_p = \tan^{-1}\left(\frac{1}{\sin\theta_c}\right) = \tan^{-1}\left(\frac{1}{\sin 34.4°}\right) = \boxed{60.5°}$$

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P38.50 For the polarizing angle,

$$\tan\theta_p = \frac{n_2}{n_1} = \frac{n}{1} = n$$

and $$\sin\theta_c = \frac{n_2}{n_1} = \frac{1}{n}$$

Thus, $$\tan\theta_p = \frac{1}{\sin\theta_c}:$$

$$\boxed{\theta_p = \tan^{-1}\left(\frac{1}{\sin\theta_c}\right) \quad \text{or} \quad \theta_p = \tan^{-1}\left(\csc\theta_c\right) \quad \text{or} \quad \theta_p = \cot^{-1}\left(\sin\theta_c\right)}$$

***P38.51** From Malus's law, the intensity of the light transmitted by the first polarizer is $I_1 = I_i \cos^2\theta_1$. The plane of polarization of this light is parallel to the axis of the first plate and is incident on the second plate. Malus's law gives the intensity transmitted by the second plate as

$$I_2 = I_1 \cos^2\left(\theta_2 - \theta_1\right) = I_i \cos^2\theta_1 \cos^2\left(\theta_2 - \theta_1\right)$$

This light is polarized parallel to the axis of the second plate and is incident upon the third plate. A final application of Malus's law gives the transmitted intensity as

$$I_f = I_2 \cos^2\left(\theta_3 - \theta_2\right) = I_i \cos^2\theta_1 \cos^2\left(\theta_2 - \theta_1\right)\cos^2\left(\theta_3 - \theta_2\right)$$

With $\theta_1 = 20.0°$, $\theta_2 = 40.0°$, and $\theta_3 = 60.0°$, this result yields

$$I_f = (10.0 \text{ units})\left(\cos^2 20.0°\right)\left(\cos^2 20.0°\right)\left(\cos^2 20.0°\right) = \boxed{6.89 \text{ units}}$$

***P38.52** Half of the unpolarized light passes through the first sheet. The light that passes through the first sheet is polarized at 45° relative to the second sheet, and the light that passes through the second sheet is polarized at 45° relative to the third sheet. The fraction of transmitted light is given by two successive applications of Malus's law:

$$\frac{I}{I_{\max}} = \frac{1}{2}\left(\cos^2 45.0°\right)\left(\cos^2 45.0°\right) = \boxed{\frac{1}{8}}$$

Additional Problems

P38.53 (a) We assume the first side maximum is at $a\sin\theta = 1.5\lambda$. (Its location is determined more precisely in Problem 71.) Then the required fractional intensity is

$$\frac{I}{I_{\max}} = \left[\frac{\sin(\pi a \sin\theta/\lambda)}{\pi a \sin\theta/\lambda}\right]^2 = \left[\frac{\sin(1.5\pi)}{1.5\pi}\right]^2 = \frac{1}{2.25\pi^2} = \boxed{0.045\ 0}$$

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(b) Proceeding as in part (a), we assume $a\sin\theta = 2.5\lambda$:

$$\frac{I}{I_{\max}} = \left[\frac{\sin(\pi a \sin\theta/\lambda)}{\pi a \sin\theta/\lambda}\right]^2 = \left[\frac{\sin(2.5\pi)}{2.5\pi}\right]^2 = \frac{1}{6.25\pi^2} = \boxed{0.016\ 2}$$

P38.54 (a) One slit, as the central maximum is twice as wide as the other maxima. A two-slit pattern has evenly spaced fringes (within a one-slit diffraction envelope).

(b) For precision, we measure from the second minimum on one side of the center to the second minimum on the other side:

$$2y = (11.7 - 6.3)\text{ cm} = 5.4\text{ cm} \rightarrow y = 2.7\text{ cm}$$

$$\tan\theta = \frac{y}{L} = \frac{0.027\text{ m}}{2.60\text{ m}} \approx \sin\theta$$

$$a\sin\theta = m\lambda$$

$$a = \frac{m\lambda}{\sin\theta} = \frac{2\left(632.8\times10^{-9}\text{ m}\right)}{\left(\frac{0.027\text{ m}}{2.60\text{ m}}\right)} = 1.22\times10^{-4}\text{ m}$$

$$= \boxed{0.122\text{ mm wide}}$$

P38.55 Figure 38.23 of the text shows the situation. This is Bragg diffraction for water waves.

$$2d\sin\theta = m\lambda \quad \text{or} \quad \lambda = \frac{2d\sin\theta}{m}$$

$m = 1$: $\lambda_1 = \dfrac{2(2.80\text{ m})\sin 80.0^\circ}{1} = \boxed{5.51\text{ m}}$

$m = 2$: $\lambda_2 = \dfrac{2(2.80\text{ m})\sin 80.0^\circ}{2} = \boxed{2.76\text{ m}}$

$m = 3$: $\lambda_3 = \dfrac{2(2.80\text{ m})\sin 80.0^\circ}{3} = \boxed{1.84\text{ m}}$

P38.56 For dark fringes in an interference pattern formed by a single slit, $a\sin\theta = m\lambda$. By the small-angle approximation, $\sin\theta \approx \tan\theta = \dfrac{y}{L}$.

Substituting, we have $a\dfrac{y}{L} = 2\lambda$ and

$$\lambda = \frac{ya}{2L} = \frac{\left(1.40\times10^{-3}\text{ m}\right)\left(0.800\times10^{-3}\text{ m}\right)}{2\left(85.0\times10^{-2}\text{ m}\right)}$$

$$= 6.59\times10^{-7}\text{ m} = \boxed{659\text{ nm}}$$

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P38.57 The first minimum is at $a\sin\theta = (1)\lambda$.

This has no solution if $\frac{\lambda}{a} > 1$,

or if $a < \lambda = \boxed{632.8\text{ nm}}$.

P38.58 (a) With light in effect moving through vacuum, Rayleigh's criterion limits the resolution according to

$$\theta_{\min} = 1.22\frac{\lambda}{D} = \frac{d}{L}$$

The diameter of the aperture is then

$$D = \frac{1.22\lambda L}{d} = \frac{1.22\left(885\times10^{-9}\text{ m}\right)(12\ 000\text{ m})}{2.30\text{ m}}$$
$$= 0.005\ 63\text{ m} = \boxed{5.63\text{ mm}}$$

(b) The assumption is unreasonable. Over a horizontal path of 12 km in air, density variations associated with convection ("heat waves," or what an astronomer calls "seeing") would make the motorcycles completely unresolvable with any optical device.

P38.59 (a) We first determine the wavelength of 1.40-GHz radio waves from $\lambda = \frac{v}{f}$:

$$\lambda = \frac{3.00\times10^{8}\text{ m/s}}{1.40\times10^{9}\text{ s}^{-1}} = 0.214\text{ m}$$

Applying Rayleigh's criterion, $\theta_{\min} = 1.22\frac{\lambda}{D}$, we obtain

$$\theta_{\min} = 1.22\left(\frac{0.214\text{ m}}{3.60\times10^{4}\text{ m}}\right) = \boxed{7.26\ \mu\text{rad}}$$

$$\theta_{\min} = (7.26\ \mu\text{rad})\left(\frac{180\times60\times60\text{ s}}{\pi}\right) = \boxed{1.50\text{ arc seconds}}$$

(b) To determine the separation between the clouds, we use $\theta_{\min} = \frac{d}{L}$:

$$d = \theta_{\min}L = \left(7.26\times10^{-6}\text{ rad}\right)(26\ 000\text{ ly}) = \boxed{0.189\text{ ly}}$$

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(c) It is not true for humans, but we assume the hawk's visual acuity is limited only by Rayleigh's criterion, $\theta_{\min} = 1.22\dfrac{\lambda}{D}$. Substituting numerical values,

$$\theta_{\min} = 1.22\left(\frac{500\times10^{-9}\ \text{m}}{12.0\times10^{-3}\ \text{m}}\right) = 50.8\ \mu\text{rad} = \boxed{10.5\ \text{seconds of arc}}$$

(d) Following the same procedure as in part (b), we have

$$d = \theta_{\min}L = \left(50.8\times10^{-6}\ \text{rad}\right)(30.0\ \text{m}) = 1.52\times10^{-3}\ \text{m} = \boxed{1.52\ \text{mm}}$$

***P38.60** Differentiating Equation 38.7, $d\sin\theta = m\lambda$, gives

$$d(\cos\theta)d\theta = m d\lambda$$

or $$d\sqrt{1-\sin^2\theta}\,\Delta\theta \approx m\Delta\lambda.$$

Plugging in for $\sin\theta$,

$$d\sqrt{1-\frac{m^2\lambda^2}{d^2}}\,\Delta\theta \approx m\Delta\lambda$$

so $$\Delta\theta \approx \frac{\Delta\lambda}{\sqrt{\left(d^2/m^2\right)-\lambda^2}}.$$

***P38.61** The grid spacing is

$$d = \frac{10^{-3}\ \text{m}}{400} = 2.50\times10^{-6}\ \text{m}$$

(a) From Equation 38.7, $d\sin\theta = m\lambda$:

$$\theta_a = \sin^{-1}\left[\frac{2\left(541\times10^{-9}\ \text{m}\right)}{2.50\times10^{-6}\ \text{m}}\right] = \boxed{25.6^\circ}$$

(b) In water,

$$\lambda = \frac{541\times10^{-9}\ \text{m}}{1.333} = 4.06\times10^{-7}\ \text{m}$$

and $$\theta_b = \sin^{-1}\left[\frac{2\left(4.06\times10^{-7}\ \text{m}\right)}{2.50\times10^{-6}\ \text{m}}\right] = \boxed{18.9^\circ}$$

(c) $d\sin\theta_a = 2\lambda$ and $d\sin\theta_b = \dfrac{2\lambda}{n} \rightarrow dn\sin\theta_b = 2\lambda$

Each equals 2λ: therefore $n\sin\theta_b = (1)\sin\theta_a$.

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P38.62 We check to see if the $m = 15$ interference maximum is visible.

We find the sine of the angle for the $m = m_{\text{double}}$ two-slit interference maximum:

$$m_{\text{double}}\lambda = d\sin\theta_{\text{bright}} \rightarrow \sin\theta_{\text{bright}} = \frac{m_{\text{double}}\lambda}{d} \quad \textbf{[1]}$$

Then find the sine of the angle for the $m = m_{\text{single}}$ single-slit interference minimum:

$$\sin\theta_{\text{dark}} = \frac{m_{\text{single}}\lambda}{a} \quad \textbf{[2]}$$

Divide equation [2] by equation [1]:

$$\frac{\sin\theta_{\text{dark}}}{\sin\theta_{\text{bright}}} = \frac{m_{\text{single}}\,\lambda/a}{m_{\text{double}}\,\lambda/d} = \frac{m_{\text{single}}}{m_{\text{double}}}\frac{d}{a}$$

Now let the angle of the single-slit minimum be equal to that of the double-slit maximum:

$$1 = \frac{m_{\text{single}}}{m_{\text{double}}}\frac{d}{a} = \frac{m_{\text{single}}}{m_{\text{double}}}\frac{30.0\ \mu m}{2.00\ \mu m} = 15\frac{m_{\text{single}}}{m_{\text{double}}}$$

which gives $m_{\text{double}} = 15m_{\text{single}}$.

Therefore, the $m_{\text{single}} = 1$ minimum aligns with the $m_{\text{double}} = 15$ maximum so that the $m_{\text{double}} = 15$ maximum has zero intensity and could not startle the co-worker.

P38.63 With a grazing angle of 36.0° (measured from the surface), the angle of incidence is 54.0°, which equals the polarizing angle:

$$\tan\theta_p = \frac{n_2}{n_1} = \frac{n}{1.00} = n = \tan 54.0° = 1.38$$

In the liquid,

$$\lambda_n = \frac{\lambda}{n} = \frac{750\text{ nm}}{1.38} = \boxed{545\text{ nm}}$$

P38.64 (a) Bragg's law applies to the space lattice of melanin rods. Consider the planes $d = 0.25\ \mu$m apart. For light at near-normal incidence, strong reflection happens for the wavelength given by $2d\sin\theta = m\lambda$. The longest wavelength reflected strongly corresponds to $m = 1$:

$$2\left(0.25\times10^{-6}\text{ m}\right)\sin 90° = \lambda = 500\text{ nm}$$

This is the blue-green color.

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(b) For light incident at grazing angle 60°, $2d\sin\theta = m\lambda$ gives $2\left(0.25\times10^{-6}\text{ m}\right)\sin 60° = \lambda = 433\text{ nm}$. This is violet.

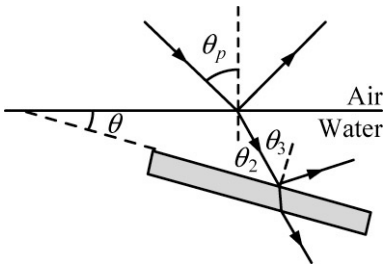
(c) Your two eyes receive light reflected from the feather at different angles, so they receive light incident at different angles and containing different colors reinforced by constructive interference.

(d) The longest wavelength that can be reflected with extra strength by these melanin rods is the one we computed first, 500 nm blue-green.

(e) If the melanin rods were farther apart (say 0.32 μm) they could reflect red with constructive interference.

P38.65 In ANS. FIG. P38.65, light strikes the liquid at the polarizing angle θ_p, enters the liquid at angle θ_2, and then strikes the slab at the angle θ_3, which is equal to the polarizing angle θ_p'. The angle between the water surface and the surface of the slab, θ, is related to the other angles by (from the triangle)

$$\theta + \left(90° + \theta_2\right) + \left(90° - \theta_3\right) = 180° \quad \rightarrow \quad \theta = \theta_3 - \theta_2$$



ANS. FIG. P38.65

For the air-to-water interface,

$$\tan\theta_p = \frac{n_{\text{water}}}{n_{\text{air}}} = \frac{1.33}{1.00} \quad \rightarrow \quad \theta_p = 53.1°$$

and $$\left(1.00\right)\sin\theta_p = \left(1.33\right)\sin\theta_2$$

$$\theta_2 = \sin^{-1}\left(\frac{\sin 53.1°}{1.33}\right) = 36.9°$$

For the water-to-slab interface,

$$\tan\theta_3 = \tan\theta_p = \frac{n_{\text{slab}}}{n_{\text{water}}} = \frac{n}{1.33} = \frac{1.62}{1.33}$$

$$\theta_3 = 50.6°$$

The angle between surfaces is $\theta = \theta_3 - \theta_2 = \boxed{13.7°}$.

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P38.66 Refer to ANS. FIG. P38.65 above. Light strikes the liquid at the polarizing angle θ_p, enters the liquid at angle θ_2, and then strikes the slab at the angle θ_3, which is equal to the polarizing angle θ_p'. The angle between the water surface and the surface of the slab, θ, is related to the other angles by (from the triangle)

$$\theta + (90° + \theta_2) + (90° - \theta_3) = 180° \quad \rightarrow \quad \theta = \theta_3 - \theta_2$$

Also,

$$\theta_p + 90° + \theta_2 = 180°$$

$$\theta_p = 90° - \theta_2$$

For the air-to-liquid interface,

$$\tan\theta_p = \frac{n_2}{n_1} = \frac{n_{\text{liquid}}}{n_{\text{air}}} = \frac{n_\ell}{1} = \frac{\sin\theta_p}{\cos\theta_p} = \frac{\sin(90° - \theta_2)}{\cos(90° - \theta_2)}$$

$$= \frac{\cos\theta_2}{\sin\theta_2} = \frac{1}{\tan\theta_2}$$

So, $$\tan\theta_2 = \frac{1}{n_\ell} \quad \rightarrow \quad \theta_2 = \tan^{-1}\left(\frac{1}{n_\ell}\right)$$

For the water-to-slab interface,

$$\tan\theta_3 = \tan\theta_p' = \frac{n_{\text{slab}}}{n_{\text{liquid}}} = \frac{n}{n_\ell} \quad \rightarrow \quad \theta_3 = \tan^{-1}\left(\frac{n}{n_\ell}\right)$$

Therefore,

$$\theta = \theta_3 - \theta_2 \quad \rightarrow \quad \boxed{\theta = \tan^{-1}\left(\frac{n}{n_\ell}\right) - \tan^{-1}\left(\frac{1}{n_\ell}\right)}$$

P38.67 For the limiting angle of resolution between lines we assume

$$\theta_{\min} = 1.22\frac{\lambda}{D} = 1.22\frac{(550\times10^{-9}\ \text{m})}{(5.00\times10^{-3}\ \text{m})} = 1.34\times10^{-4}\ \text{rad}$$

Assuming a picture screen with vertical dimension ℓ, the minimum viewing distance for no visible lines is found from $\theta_{\min} = \dfrac{\ell/485}{L}$. The desired ratio is then

$$\frac{L}{\ell} = \frac{1}{485\theta_{\min}} = \frac{1}{485(1.34\times10^{-4}\ \text{rad})} = \boxed{15.4}$$

When the pupil of a human eye is wide open, its actual resolving power is significantly poorer than Rayleigh's criterion suggests.

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P38.68 (a) We require

$$\theta_{\min} = 1.22\frac{\lambda}{D} = \frac{\text{radius of diffraction disk}}{L} = \frac{D/2}{L}.$$

Then, $\boxed{D^2 = 2.44\lambda L}$.

(b) $D = \sqrt{2.44\left(500\times10^{-9}\text{ m}\right)\left(0.150\text{ m}\right)} = 4.28\times10^{-4}\text{ m} = \boxed{428\ \mu\text{m}}$

P38.69 (a) Constructive interference of light of wavelength λ on the screen is described by $d\sin\theta = m\lambda$ and, because $\tan\theta = \frac{y}{L}$, we may write $\sin\theta = \frac{y}{\sqrt{L^2+y^2}}$. Therefore,

$$(d)y\left(L^2+y^2\right)^{-1/2} = m\lambda$$

Differentiating with respect to y gives

$$(d)\left(L^2+y^2\right)^{-1/2} + (d)y\left(-\frac{1}{2}\right)\left(L^2+y^2\right)^{-3/2}\left(0+2y\right) = m\frac{d\lambda}{dy}$$

$$\frac{(d)}{\left(L^2+y^2\right)^{1/2}} - \frac{(d)y^2}{\left(L^2+y^2\right)^{3/2}} = m\frac{d\lambda}{dy} = \frac{(d)\left(L^2+y^2\right)-(d)y^2}{\left(L^2+y^2\right)^{3/2}}$$

$$\rightarrow\ \frac{d\lambda}{dy} = \frac{(d)L^2}{m\left(L^2+y^2\right)^{3/2}}$$

(b) Here $d\sin\theta = m\lambda$ gives, for $m = 1$,

$$\frac{10^{-2}\text{ m}}{8\ 000}\sin\theta = 1\left(550\times10^{-9}\text{ m}\right)$$

or $$\theta = \sin^{-1}\left(\frac{550\times10^{-9}\text{ m}}{1.25\times10^{-6}\text{ m}}\right) = 26.1°$$

Then,

$$y = L\tan\theta = (2.40\text{ m})\tan 26.1° = 1.18\text{ m}$$

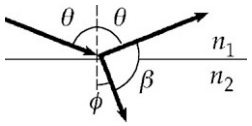
So we have

$$\frac{d\lambda}{dy} = \frac{(d)L^2}{m\left(L^2+y^2\right)^{3/2}} = \frac{\left(1.25\times10^{-6}\text{ m}\right)(2.40\text{ m})^2}{(1)\left[(2.4\text{ m})^2+(1.18\text{ m})^2\right]^{3/2}}$$

$$= 3.77\times10^{-7}\frac{\text{m}}{\text{m}} = 3.77\times10^{-7}\frac{10^9\text{nm}}{10^2\text{cm}} = \boxed{3.77\ \text{nm/cm}}$$

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P38.70 (a) Applying Snell's law gives $n_2 \sin\phi = n_1 \sin\theta$. From the sketch in ANS. FIG. P38.70(a), we also see that:



ANS. FIG. P38.70(a)

$$\theta + \phi + \beta = \pi, \quad \text{or} \quad \phi = \pi - (\theta + \beta)$$

Using the given identity,

$$\sin\phi = \sin\pi\cos(\theta+\beta) - \cos\pi\sin(\theta+\beta)$$

which reduces to,

$$\sin\phi = \sin(\theta+\beta)$$

Applying the identity again,

$$\sin\phi = \sin\theta\cos\beta + \cos\theta\sin\beta$$

Snell's law then becomes,

$$n_2(\sin\theta\cos\beta + \cos\theta\sin\beta) = n_1\sin\theta$$

or (after dividing by $\cos\theta$):

$$n_2(\tan\theta\cos\beta + \sin\beta) = n_1\tan\theta$$

Solving for $\tan\theta$ gives:

$$\boxed{\tan\theta = \frac{n_2\sin\beta}{n_1 - n_2\cos\beta}}$$

(b) If $\beta = 90.0°$, the above result becomes:

$$\tan\theta = \frac{n_2\sin 90°}{n_1 - n_2\cos 90°} = \frac{n_2}{n_1}, \text{ which is Brewster's law}$$

P38.71 From $I = I_{\max}\left(\frac{\sin\phi}{\phi}\right)^2$ we find

$$\frac{dI}{d\phi} = I_{\max} 2\left(\frac{\sin\phi}{\phi}\right)\left(\frac{\phi\cos\phi - [\sin\phi]1}{\phi^2}\right)$$

and require that it be zero. The possibility $\sin\phi = 0$ locates all of the minima and the central maximum, according to

$$\phi = 0,\ \pi,\ 2\pi,\ \ldots; \qquad \phi = \frac{\pi a\sin\theta}{\lambda} = 0,\ \pi,\ 2\pi,\ \ldots;$$

$$a\sin\theta = 0,\ \lambda,\ 2\lambda,\ \ldots$$

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The side maxima are found from

$$\phi\cos\phi - \sin\phi = 0 \quad \text{or} \quad \tan\phi = \phi$$

This has solutions

$$\phi = 4.493\,4,\ \phi = 7.7253, \text{ and others.}$$

(a) $\boxed{\phi = 4.49 \text{ compared to the prediction from the approximation of } 1.5\pi = 4.71.}$

(b) $\boxed{\phi = 7.73 \text{ compared to the prediction from the approximation of } 2.5\pi = 7.85.}$

P38.72 (a) From Equation 38.2, $\dfrac{I}{I_{\max}} = \left[\dfrac{\sin(\phi)}{\phi}\right]^2$ where we define

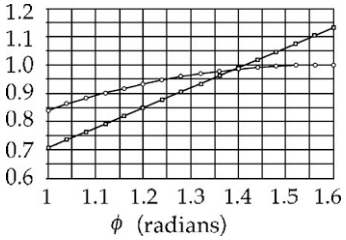
$$\phi \equiv \frac{\pi a \sin\theta}{\lambda}.$$

Therefore, when $\dfrac{I}{I_{\max}} = \dfrac{1}{2}$ we must have

$$\frac{\sin\phi}{\phi} = \frac{1}{\sqrt{2}}, \quad \text{or} \quad \boxed{\sin\phi = \frac{\phi}{\sqrt{2}}}$$

(b) Let $y_1 = \sin\phi$ and $y_2 = \dfrac{\phi}{\sqrt{2}}$.

A plot of y_1 and y_2 in the range $1.00 \le \phi \le \dfrac{\pi}{2}$ is shown in ANS. FIG. P38.72(b).



ANS. FIG. P38.72(b)

The solution to the transcendental equation is found to be $\boxed{\phi = 1.39 \text{ rad}}$.

(c) $\dfrac{\pi a \sin\theta}{\lambda} = \phi$ gives $\sin\theta = \left(\dfrac{\phi}{\pi}\right)\dfrac{\lambda}{a}$. If $\dfrac{\lambda}{a}$ is small, then $\theta \approx \left(\dfrac{\phi}{\pi}\right)\dfrac{\lambda}{a}$.

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This gives the half-width, measured away from the maximum at $\theta = 0$. The pattern is symmetric, so the full width is given by

$$\Delta\theta = \left(\frac{\phi}{\pi}\right)\frac{\lambda}{a} - \left(-\left(\frac{\phi}{\pi}\right)\frac{\lambda}{a}\right) = 2\left(\frac{\phi}{\pi}\right)\frac{\lambda}{a} = 2\left(\frac{1.39 \text{ rad}}{\pi}\right)\frac{\lambda}{a}$$

$$= \boxed{\frac{0.885\lambda}{a}}$$

(d)

ϕ	$\sqrt{2}\sin\phi$	
1	1.19	bigger than ϕ
2	1.29	smaller than ϕ
1.5	1.41	smaller
1.4	1.394	
1.39	1.391	bigger
1.395	1.392	
1.392	1.391 7	smaller
1.391 5	1.391 54	bigger
1.391 52	1.391 55	bigger
1.391 6	1.391 568	smaller
1.391 58	1.391 563	
1.391 57	1.391 561	
1.391 56	1.391 558	
1.391 559	1.391 557 8	
1.391 558	1.391 557 5	
1.391 557	1.391 557 3	
1.391 557 4	1.391 557 4	

We get the answer as 1.391 557 4 to seven digits after 17 steps. Clever guessing, like using the value of $\sqrt{2}\sin\phi$ as the next guess for ϕ, could reduce this to around 13 steps.

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P38.73 (a) The angles of bright beams diffracted from the grating are given by $d\sin\theta = m\lambda$. The angular dispersion is defined as the derivative $\frac{d\theta}{d\lambda}$:

$$d\cos\theta\frac{d\theta}{d\lambda} = m \quad \rightarrow \quad \frac{d\theta}{d\lambda} = \frac{m}{d\cos\theta}$$

(b) For the average wavelength

$$\frac{579.065\text{ nm} + 576.959\text{ nm}}{2} = 578.012\text{ nm}$$

$d\sin\theta = m\lambda$ gives

$$\frac{0.020\,0\text{ m}}{8\,000}\sin\theta = 2\left(578.012\times10^{-9}\text{ m}\right)$$

and $\theta = \sin^{-1}\dfrac{2\times578\times10^{-9}\text{ m}}{2.5\times10^{-6}\text{ m}} = 27.5°$

The separation angle between the lines is, for

$$\Delta\lambda = 576.959\text{ nm} - 579.065\text{ nm} = 2.106\text{ nm}$$

and

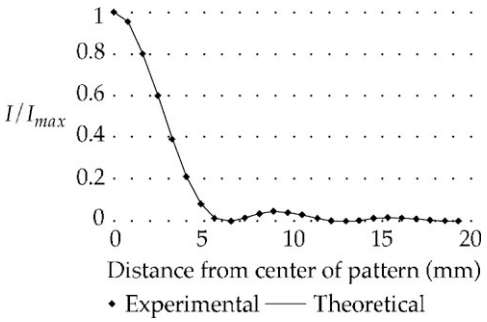
$$\Delta\theta = \frac{d\theta}{d\lambda}\Delta\lambda = \frac{m}{d\cos\theta}\Delta\lambda$$

$$= \frac{2}{2.5\times10^{-6}\text{ m}\cos27.5°}\left(2.106\times10^{-9}\text{ m}\right)$$

$$= 0.001\,90 = 0.001\,90\text{ rad} = 0.001\,90\text{ rad}\left(\frac{180°}{\pi\text{ rad}}\right)$$

$$= \boxed{0.109°}$$

P38.74 (a) See ANS. FIG. P38.74.



ANS. FIG. P38.74

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(b) The first minimum in the single-slit diffraction pattern occurs at

$$\sin\theta = \frac{\lambda}{a} \approx \frac{y_{\min}}{L}$$

Thus, the slit width is given by

$$a = \frac{\lambda L}{y_{\min}}$$

For a minimum located at $y_{\min} = 6.36 \text{ mm} \pm 0.08 \text{ mm}$, the width is

$$a = \frac{(632.8\times10^{-9} \text{ m})(1.00 \text{ m})}{6.36\times10^{-3} \text{ m}} = \boxed{99.5\ \mu\text{m} \pm 1\%}$$

Challenge Problems

P38.75 (a) The *E* and *O* rays, in phase at the surface of the plate, will have a phase difference

$$\theta = \left(\frac{2\pi}{\lambda}\right)\delta$$

after traveling distance d through the plate. Here δ is the difference in the *optical path* lengths of these rays. The optical path length between two points is the product of the actual path length d and the index of refraction. Therefore,

$$\delta = |dn_O - dn_E|$$

The absolute value is used since $\frac{n_O}{n_E}$ may be more or less than unity. Therefore,

$$\theta = \left(\frac{2\pi}{\lambda}\right)|dn_O - dn_E| = \boxed{\left(\frac{2\pi}{\lambda}\right)d|n_O - n_E|}$$

(b) $$d = \frac{\lambda\theta}{2\pi|n_O - n_E|} = \frac{(550\times10^{-9} \text{ m})(\pi/2)}{2\pi|1.544 - 1.553|} = 1.53\times10^{-5} \text{ m} = \boxed{15.3\ \mu\text{m}}$$

P38.76 (a) The concave mirror of the spy satellite is probably about 2 m in diameter, and is surely not more than 5 m in diameter. That is the size of the largest piece of glass successfully cast to a precise shape, for the mirror of the Hale telescope on Mount Palomar. If the spy satellite had a larger mirror, its manufacture could not be kept secret, and it would be visible from the ground. Outer space

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is probably closer than your state capitol, but the satellite is surely above 200-km altitude, for reasonably low air friction. We find the distance between barely resolvable objects at a distance of 200 km, seen in yellow light through a 5-m aperture:

$$\frac{y}{L} = \theta_{\min} = 1.22\frac{\lambda}{D}$$

$$y = \left(2\times10^{7}\ \text{m}\right)(1.22)\left(\frac{6\times10^{-7}\ \text{m}}{5\ \text{m}}\right) = 3\ \text{cm}$$

Thus the snooping spy satellite cannot see the difference between III and II or IV on a license plate. A resolution of about 3 cm would make it difficult to read a license plate.

(b) No. The resolution is too large. It cannot count coins spilled on a sidewalk, much less read the dates on them.

Considering atmospheric image distortion caused by variations in air density and temperature, the distance between barely resolvable objects is more like, assuming a limiting angle of one second of arc,

$$\left(2\times10^{7}\ \text{m}\right)(1\ \text{s})\left(\frac{1^\circ}{3\,600\ \text{s}}\right)\left(\frac{\pi\ \text{rad}}{180^\circ}\right) = 97\ \text{cm} \approx 1\ \text{m}$$

P38.77 (a) From Equation 38.1, $\theta = \sin^{-1}\left(\frac{m\lambda}{a}\right)$. In this case $m = 1$ and

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{7.50\times10^{9}\ \text{Hz}} = 4.00\times10^{-2}\ \text{m}$$

Thus,

$$\theta = \sin^{-1}\left(\frac{4.00\times10^{-2}\ \text{m}}{6.00\times10^{-2}\ \text{m}}\right) = \boxed{41.8^\circ}$$

(b) From Equation 38.2,

$$\frac{I}{I_{\max}} = \left[\frac{\sin(\phi)}{\phi}\right]^2 \quad \text{where} \quad \phi = \frac{\pi a \sin\theta}{\lambda}$$

When $\theta = 15.0^\circ$,

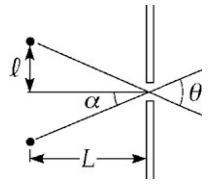
$$\phi = \frac{\pi\left(0.060\,0\ \text{m}\right)\sin 15.0^\circ}{0.040\,0\ \text{m}} = 1.22\ \text{rad}$$

and $$\frac{I}{I_{\max}} = \left[\frac{\sin(1.22\ \text{rad})}{1.22\ \text{rad}}\right]^2 = \boxed{0.592}$$

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(c) $\sin\theta = \dfrac{\lambda}{a}$, so $\theta = 41.8°$:

This is the minimum angle subtended by the two sources at the slit. Refer to ANS. FIG. P38.77(c). Let α be the half angle between the sources, each a distance $\ell = 0.100$ m from the center line and a distance L from the slit plane. Then,



ANS. FIG. P38.77(c)

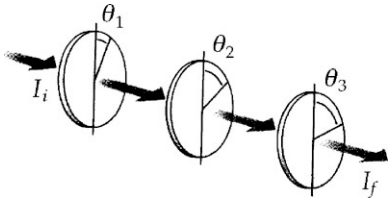
$$L = \ell\cot\alpha = (0.100\text{ m})\cot\left(\frac{41.8°}{2}\right) = \boxed{0.262\text{ m}}$$

P38.78 For incident unpolarized light of intensity I_{max}, the average value of the cosine-squared function is one-half, so the intensity after transmission by the first disk is $I = \dfrac{1}{2}I_{max}$.

After transmitting 2nd disk: $I = \dfrac{1}{2}I_{max}\cos^2\theta$

After transmitting 3rd disk: $I = \dfrac{1}{2}I_{max}\cos^2\theta\cos^2(90° - \theta)$

where the angle between the first and second disk is $\theta = \omega t$.



ANS. FIG. P38.78

Using trigonometric identities $\cos^2\theta = \dfrac{1}{2}(1 + \cos 2\theta)$

and $\cos^2(90° - \theta) = \sin^2\theta = \dfrac{1}{2}(1 - \cos 2\theta)$,

we have $I = \dfrac{1}{2}I_{max}\left[\dfrac{(1 + \cos 2\theta)}{2}\right]\left[\dfrac{(1 - \cos 2\theta)}{2}\right]$

$$I = \frac{1}{8}I_{max}\left(1 - \cos^2 2\theta\right) = \frac{1}{8}I_{max}\left(\frac{1}{2}\right)(1 - \cos 4\theta)$$

Since $\theta = \omega t$, the intensity of the emerging beam is given by

$$\boxed{I = \frac{1}{16}I_{max}\left(1 - \cos\ 4\omega t\right)}$$

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P38.79 The energy in the central maximum we can estimate in Figure P38.79 as proportional to

$$(\text{width})(\text{height}) = (2\pi) I_{\max}$$

As in Problem P38.71, the maximum height of the first side maximum is approximately

$$I = I_{\max}\left[\frac{\sin(\phi)}{\phi}\right]^2 = I_{\max}\left[\frac{\sin(3\pi/2)}{3\pi/2}\right]^2 = \frac{4I_{\max}}{9\pi^2}$$

Then the energy in one side maximum is proportional to $\pi\left(\frac{4I_{\max}}{9\pi^2}\right)$, and that in both of the first side maxima together is proportional to $2\pi\left(\frac{4I_{\max}}{9\pi^2}\right)$.

Similarly and more precisely, and always with the same proportionality constant, the energy in both of the second side maxima is proportional to $2\pi\left(\frac{4I_{\max}}{25\pi^2}\right)$.

The energy in all of the side maxima together is proportional to

$$2\pi\left(\frac{4I_{\max}}{\pi^2}\right)\left(\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}+\frac{1}{9^2}+\cdots\right)$$

$$= 2\pi\left(\frac{4I_{\max}}{\pi^2}\right)\left(\frac{\pi^2}{8}-1\right) = I_{\max}\left(\pi - \frac{8}{\pi}\right) = 0.595 I_{\max}$$

The ratio of the energy in the central maximum to the total energy is then

$$\frac{(2\pi) I_{\max}}{(2\pi) I_{\max} + 0.595 I_{\max}} = \frac{1}{1 + 0.595/(2\pi)} = 0.913 = 91.3\%$$

Our calculation is only a rough estimate, because the shape of the central maximum in particular is not just a vertically-stretched cycle of a cosine curve. It is slimmer than that.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P38.2 4.22 mm

P38.4 (a) 51.8 μm; (b) 949 μm; (c) horizontal; (d) vertical; (e) A smaller distance between aperture edges causes a wider diffraction angle. The longer dimension of each rectangle is 18.3 times larger than the smaller dimension.

P38.6 (a) 1.50 m; (b) 4.05 mm

P38.8 $a = \dfrac{\lambda L \left| m_2 - m_1 \right|}{\Delta y}$

P38.10 See P38.10 for full explanation.

P38.12 (a) $\theta_0 = 0°$, $\theta_1 = \pm 10.3°$, $\theta_2 = \pm 21.0$, $\theta_3 = \pm 32.5°$, $\theta_4 = \pm 45.8°$, $\theta_5 = \pm 63.6°$; (b) 11, (c) $\theta_1 = \pm 45.8°$, (d) two, (e) two, (f) nine, (g) $0.032\,4\, I_{\max}$

P38.14 1.00×10^{-3} rad

P38.16 25.0 m

P38.18 (a) 79.8 μrad; (b) violet, 54.2 μrad; (c) The resolving power is improved, with the minimum resolvable angle becoming 60.0 μrad.

P38.20 3.09 m

P38.22 (a) Blue; (b) 186 m to 271 m

P38.24 105 m

P38.26 7.35°

P38.28 (a) 479 nm, 647 nm, 698 nm; (b) 20.5°, 28.3°, 30.7°

P38.30 (a) 5 orders is the maximum; (b) 10 orders in the short-wavelength region

P38.32 5.91°, 13.2°, 26.5°

P38.34 $\theta_{2r} > \theta_{3v}$ and these orders must overlap.

P38.36 (a) 0.738 mm; (b) See P38.36(b) for full explanation.

P38.48 $\theta = 14.4°$

P38.40 $\theta = 31.9°$

P38.42 See P38.42 for full explanation.

P38.44 1.11

P38.46 (a) 93.3%; (b) 50.0%; (c) 0.00%

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P38.48 (a) 0.875; (b) 0.789; (c) 0.670; (d) We can get more and more of the incident light through the stack of ideal filters, approaching 50%, by reducing the angle between the transmission axes of each one and the next.

P38.50 $\theta_p = \tan^{-1}\left(\dfrac{1}{\sin\theta_c}\right)$ or $\theta_p = \tan^{-1}\left(\csc\theta_c\right)$ or $\theta_p = \cot^{-1}\left(\sin\theta_c\right)$

P38.52 1/8

P38.54 (a) One slit, as the central maximum is twice as wide as the other maxima; (b) 0.122 mm wide

P38.56 659 nm

P38.58 (a) 5.63 mm; (b) The assumption is unreasonable. Over a horizontal path of 12 km in air, density variation associated with convection would make the motorcycles completely unresolvable with any optical device.

P38.60 See P38.60 for full explanation.

P38.62 See P38.62 for full explanation.

P38.64 (a–e) See P38.64 for full explanations.

P38.66 $\theta = \tan^{-1}\left(\dfrac{n}{n_\ell}\right) - \tan^{-1}\left(\dfrac{1}{n_\ell}\right)$

P38.68 (a) $D^2 = 2.44\lambda L$; (b) 428 μm

P38.70 (a) $\tan\theta = \dfrac{n_2 \sin\beta}{n_1 - n_2\cos\beta}$; (b) See P38.70(b) for full explanation.

P38.72 (a) $\sin\phi = \dfrac{\phi}{\sqrt{2}}$; (b) $\phi = 1.39$ rad; (c) $\dfrac{0.885\lambda}{a}$; (d) 17 steps (13 with clever guessing)

P38.74 (a) See ANS FIG P38.74; (b) 99.5 μm $\pm$ 1%

P38.76 (a) A resolution of about 3 cm would make it difficult to read a license plate; (b) No

P38.78 $\dfrac{1}{16} I_{\max}\left(1 - \cos 4\omega t\right)$

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